

Multiple Streams, Recoil-Tactics Policy Entrepreneurship and the New-Look EU HTA Framework

Vassilis Karokis-Mavrikos (University of Surrey)

ECPR General Conference 2025

Panel: Using the Multiple Streams Framework to Explain Policy Change and
Stasis

Abstract

Health Technology Assessment (HTA) determines the added value of new health technologies to inform pricing and reimbursement decisions, relying on a range of medical, economic, social, ethical, legal and other criteria. In the EU, despite two decades of voluntary collaborative initiatives, HTA had historically remained within the remit of the Member States. Nevertheless, in 2018, a surprising and ambitious Commission proposal culminated in the passing of Regulation (EU) 2021/2282. From January 2025, a new EU HTA framework comes in operation, including expansive Joint Clinical Assessments, Joint Scientific Consultations, and the establishment of a permanent governing body, the “Coordination Group”. This paper deploys a modified iteration of the Multiple Streams Framework – the EU-MSF – to determine how structural policy change for HTA emerged after decades of only incremental progression. The analysis relies on template-based document analysis and elite interviews with 40 elite stakeholders. The conclusions of the study illuminate the impact of policy networks and strategic policy entrepreneurship by the European Commission on policy change. The paper introduces the strategy of “Recoil Tactics” policy entrepreneurship to the literature and makes contributions to EU public policy and pharmaceutical policy.

Introduction:

Health Technology Assessment (HTA) is a multidisciplinary process that evaluates the added value of new health technologies – such as medicinal products, medical equipment, and diagnostic, treatment, rehabilitation, and prevention methods – compared to existing alternatives. It also examines the cost implications of these technologies for patients, as well as their impact on healthcare system organisation and treatment administration. In doing so, HTA considers the medical, economic, organisational, social, and ethical dimensions of introducing new health technologies to the market (O’Rourke et al., 2020; Kristensen et al., 2017; Gutiérrez-Ibarluzea et al., 2017).

HTA was introduced in Europe in the late 1980s (Banta, 2003), and by the 2010s it had become an integral part of pricing and reimbursement decisions, particularly for medicinal products (Allen et al., 2017). Once market authorisation is granted by the European Medicines Agency (EMA) through a centralised process confirming that new medicines are safe for circulation within the EU Single Market, HTA evaluations conducted by dedicated Member State agencies inform the decisions of national pricing and reimbursement authorities regarding the product’s launch in each Member State market.

Throughout the three decades of HTA’s expansion in Europe, EU policymaking in the field remained incremental. Between 1993 and 2021, a series of collaborative projects were launched at the initiative of a growing number of HTA agencies – initially under the auspices of the European Commission and, since 2009, with the additional support of Member States through Joint Action (JA) status. Yet despite growing participation and an expanding scope, collaboration remained largely voluntary, with no institutionalised obligations or centralised mechanisms to adopt joint outputs or integrate them into national policy processes.

Nevertheless, in 2018 the European Commission put forward a disruptive proposal aimed at fundamentally transforming the HTA landscape in the EU (European Commission, 2018a). By 2021, EU policymakers had adopted a new EU HTA framework, which came into effect in January 2025, mandating Member State participation in Joint Clinical Assessments (JCAs) for all centrally authorised medicinal products and most categories of medical devices, with implementation to be phased in over time (Regulation (EU) 2021/2282). Additionally, it establishes wide-ranging Joint Scientific Consultations (JSCs), and creates a permanent governing body – the Coordination Group – composed of national HTA representatives from all 27 EU Member States.

Regulation (EU) 2021/2282 marks one of the most recent instances of structural policy change in the EU – a shift in both the goals and means of an established policy paradigm (Hall, 1993), after decades of incrementalism. To this end, it prompts investigating the drivers of agenda setting and policy reform within a political system that has been traditionally under-studied by policy process scholarship.

Currently experiencing its longest period of “constitutional stability” – i.e., without a Treaty revision – since the era of Eurosclerosis (Dinan, 2008; Giersch, 1985), the EU has evolved into a political system without a state (Hix & Høyland, 2011), characterized by a unique multi-level governance architecture (Hooghe &

Marks, 2009). Owing largely to this *sui generis* character, EU policy developments have traditionally been analysed through macro-oriented frameworks that emphasize the integration/disintegration dichotomy – namely, the shifting delegation of powers to and from the supranational level (Hooghe & Marks, 2019). While this approach has proven highly effective in explaining key milestones in the EU’s “constitutional” evolution (i.e., Treaty revisions), contemporary policy change – occurring primarily through the Ordinary Legislative Procedure (OLP) – calls for a different lens. A meso-level perspective, which highlights how ideas, interests, and institutions interact to shape transformative policy outputs, offers a better means of capturing these dynamics (Weible, 2023; Hall, 1997). To this end, translating the tools of policy process scholarship to the EU level remains a pressing challenge today.

This paper contributes to the emerging EU policy process research agenda by addressing the question: *Why was an expansive EU HTA framework institutionalised in 2021, after more than three decades of only incremental developments in HTA collaboration at the EU level?* To answer this, it applies an adapted iteration of the Multiple Streams Framework (MSF) tailored to the EU context – the EU-MSF (Karokis-Mavrikos, 2025). Drawing on original primary data, the paper links the level of policy network integration to the mechanisms of attention mobilisation and the nature of successful policy alternatives, testing and confirming two hypotheses:

H1: In weakly integrated EU policy networks, achieving attention mobilisation requires the activation of horizontal attention-mobilisation mechanisms, i.e., those that communicate problems across EU-level policy venues and stakeholders.

H2: In weakly integrated EU policy networks, successful policy alternatives depend more heavily on technical feasibility than on value acceptability.

The study further concludes that the emergence and adoption of the new transformative EU HTA framework was driven by the Commission’s successful policy entrepreneurship, pursued through what this study labels *recoil-tactics*: a strategy of identifying a range of technically feasible alternatives under conditions of ambiguity and proposing the most preferred option within this range; confident that ensuing deliberation will still yield transformative policy change despite pushback.

Therefore, the study’s contributions are summarised as follows: First, it presents one of the first accounts in the literature explaining the emergence of the new EU HTA framework. Second, it highlights the strength of the EU-MSF as a policy process toolkit capable of guiding evaluations of the drivers of policy change in the EU context. Third, it contributes to the development of a systematic EU-MSF research programme, grounded in testable hypotheses. Finally, it introduces the concept of *recoil-tactics*, advancing the understanding of policy entrepreneurship by EU-level actors and underscoring the impact of EU policy process particularities – such as public consultations, roadmaps, and impact assessments – on the strategic choices of policy entrepreneurs.

The following sections first reflect on the study of EU policy change before introducing the EU-MSF and presenting the research design. The analysis then follows the structure of the three streams, showing how

attention mobilisation on HTA through horizontal mechanisms since 2016 was coupled with an ambitious yet widely regarded as technically feasible policy proposal, alongside the convergence of political narratives, driven by the Commission's deployment of *recoil-tactics* policy entrepreneurship. Finally, the conclusion summarises the findings and delineates the study's contributions to the literature.

Studying EU-level Policy Change

Traditionally, the scholarship on EU-level policy outputs has primarily focused on the Union's supranational governance architecture. Neofunctionalist accounts (Schmitter, 2005; Haas, 1958) attribute the transfer of competences to EU institutions across sectors to consecutive spillovers, driven by collective economic and transactional gains. In contrast, Liberal Intergovernmentalism explains EU-level stability and stasis through the calculated, collective decision-making of rationally motivated Member States (Moravcsik, 1998). Federalist, historical institutionalist, and constructivist accounts emphasise the roles of political leadership, critical junctures, and normative feedback loops, respectively (Risse, 2009; Burgess, 2000; Pierson, 1996). Most recently, postfunctionalism has highlighted identity perceptions and national party characteristics as key drivers of integration and disintegration within the EU's multi-level governance system (Hooghe & Marks, 2009; Marks et al., 1996).

While this body of scholarship has been successful in explaining the EU's macro-political evolution across successive Treaty revisions, it has limited analytical capacity in capturing the drivers of structural policy reforms within the EU's current institutional landscape. In recent years, particularly since the Treaty of Lisbon (2009), scholars have increasingly converged in viewing the EU as a distinct political system (Hix & Høyland, 2011). As a result, shifting the analytical focus from the macro to the meso level, alongside a greater emphasis on policy process dynamics, provides a more comprehensive foundation for understanding EU-level policy change – much like in national political systems (Zahariadis, 2013).

To this end, scholars have sought to extend and adapt policy process frameworks to the EU context (e.g. von Malmberg, 2023; Brooks, 2018; Benson & Russell, 2015; Princen, 2013; Citi, 2013). Among these efforts, the literature on the Multiple Streams Framework (MSF) and policy entrepreneurship arguably constitutes the most promising body of research to date. The MSF draws on the 'garbage can model' of organisational choice (Cohen, March and Olsen, 1972) to study policymaking under conditions of ambiguity – that is when multiple ways of interpreting the same circumstances exist (Zahariadis et al., 2023; Kingdon, 1984). The EU's political system has repeatedly been characterised as “an emerging garbage can” (Richardson, 2001), “an obvious candidate” for garbage can analysis (Olsen, 2001), and a “loosely coupled system” (Weick, 2001, in Zahariadis, 2007), exhibiting fluidity in stakeholder participation, opacity in organisational capacity, dynamic actor preferences, and high procedural complexity. As the MSF operationalizes institutions only implicitly – thus reducing contextual bias and encouraging transferability (Cairney and Heikkilä, 2014) – the EU provides a highly suitable setting for MSF research.

Zahariadis (2008), Ackrill et al. (2013), and Exadaktylos (2023) have emphasised the theoretical transferability of the MSF to the EU level, while Borrás and Radaelli (2011), Bache (2012), and Goyal et al. (2021) have applied the MSF to quality-of-life policymaking, the emergence of the Lisbon Strategy, and the adoption of the EU General Data Protection Regulation, respectively. Similarly, Braun (2009), Palmer (2014)

and Thierse (2017) have explored policy entrepreneurship in environmental policy case studies, focusing on both institutional and non-institutional actors.

While these studies have consistently highlighted the need for contextually-driven adaptations in the analytical tools and expectations of both the MSF and the policy entrepreneurship literature to accommodate the particularities of the EU policy process, a systematic programme of theory building and testing has yet to emerge. Herweg (2016), offers one of the most comprehensive efforts to evaluate the appropriateness of the MSF's tools in the EU context, focusing on European gas regulation. Building on this research strategy, Karokis-Mavrikos (2025) has introduced the 'EU-MSF,' a comprehensive adaptation of the framework to the modern configuration of the EU political system.

The following section summarises the key theoretical propositions of the EU-MSF, which provides the theoretical foundation for this study.

The EU-MSF

The EU-MSF retains the MSF's core hypothesis: *policy outcomes result from policy entrepreneurs successfully exploiting windows of opportunity to couple three independent and ever-flowing streams – problems, policies, and politics – through strategic action*. Nevertheless, necessary adaptations are introduced where the functional equivalents of the MSF's analytical components cannot be identified in the EU political system.

The Problem Stream

The problem stream has consistently been regarded as the most readily adaptable to the EU context (Herweg, 2017; Ackrill et al., 2013; Zahariadis, 2008). According to the MSF, three types of mechanisms – *alarming deteriorations in indicators, policy feedback, and focusing events* – can mobilize policymakers' attention to emerging problems. While all three are well-documented within the EU political system, the multi-level governance architecture introduces an additional layer of complexity by making the direction of attention mobilization an important factor in assessing the problem stream's ripeness.

As such, in the case of *indicators* and *policy feedback*, the EU-MSF identifies *upward vertical* mechanisms, which operate from the national to the EU level; *downward vertical* mechanisms, which function from the international to the EU level; and *horizontal* mechanisms, which emerge across venues or policy subsystems at the EU level.

On the one hand, the direction of attention mobilisation reveals the origins of dissatisfaction and the pathways through which it is disseminated. On the other hand, while not the explicit focus of this study, it can also shape the nature of the problems ultimately identified. For example, upward policy feedback, voluntarily communicated by Member States to EU policy venues such as the European Commission, may be motivated by blame-shifting, as national governments seek to downplay their own failures. By contrast, horizontal ex-post policy feedback mechanisms – initiated by the competent EU authorities with the participation of relevant EU-level stakeholders – are more likely to raise issues of compliance and unintended consequences (Smismans, 2015; Stern, 2009).

Finally, adapting the definition of focusing events to the EU level (e.g., Birkland et al., 2022; Birkland, 1998), they can be understood as *unpredictable shocks with harmful effects on the functioning of polities, which may originate at the national or international level but are abruptly and simultaneously recognised by most participants within one or more EU policy subsystems*.

The Politics Stream

In contrast to the problem stream, the politics stream has been identified as the one requiring the most far-reaching adaptations (Herweg, 2016). The politics stream conceptualizes the political willingness and determination necessary for policy change (Kingdon, 1984). In national-level MSF accounts, it is shaped by *legislative or administrative turnover, organized interest group action*, and the *national mood* (Herweg et al., 2018). However, in the EU context, the impact of *legislative turnover* remains debated, as decision-making in the European Parliament (EP) frequently relies on ad hoc coalitions that vote together on most issues (Herweg, 2016; Decker & Sonnicksen, 2011), and the existence of a ‘European mood’ is also contested (Pannico, 2020; Beyers et al., 2018; Herweg, 2016).

As such, the EU-MSF proposes that evaluating the ripeness of the politics stream is best achieved through an assessment of *political narratives*, as communicated by the three EU institutions participating in the Ordinary Legislative Procedure (OLP): the Council of the European Union, the European Parliament, and the European Commission. Drawing on insights from European integration scholarship, the EU-MSF posits that the discourse of each of these three institutions aligns with a distinct perception of the EU public.

Specifically, the Commission is expected to serve a ‘*functional*’ perception of the EU public, prioritizing technocratic expertise, process simplification, and the reduction of transactional costs. The EP, by contrast, is expected to embody a ‘*popular*’ perception of the EU public, emphasizing democratic accountability and the mitigation of social, political, and economic inequalities. Finally, the Council is expected to embody a ‘*ruling*’ perception of the EU public, conflating national public preferences with national interests as articulated by government officials and prioritising national strengthening through monetary and regulatory benefits. In this way, political willingness or determination for EU-level policy change is evaluated through the degree of compatibility between policy alternatives and institutional narratives.

Like in national settings, *windows of opportunity* in the EU-MSF may open in either the problem or the politics stream, driven by the respective stream mechanisms, particularly focusing events and shifts in political narratives.

The Policy Stream

Finally, the policy stream captures the emergence of transformative policy alternatives and provides the primary operating space for policy entrepreneurs. MSF scholarship identifies *technical feasibility*, *value acceptability* and *resource adequacy* as essential criteria for reformative ideas to survive and gain traction. While early MSF accounts (Kingdon, 1984) proposed an omnipresent softening process until ideas sufficiently met the three criteria, modern MSF scholarship has connected the nature of emerging policy alternatives to the levels of policy network integration (Novotný et al., 2020; Herweg et al., 2018; Zahariadis, 2003).

Policy networks are defined as “constellations of actors and their interaction in a policy sector” (Zahariadis and Allen, 1995). Drawing on the relevant literature (see Rhodes and Marsh, 1992; Rhodes, 1990; Jordan, 1990), policy networks within policy process research are typically classified as highly or weakly integrated, based on *size*, *ease of access*, *administrative capacity* and the *mode of interaction between participating actors*.

Highly integrated networks are typically smaller in size, have high barriers to entry for new stakeholders, are defined by well-embedded institutional and procedural infrastructure (high administrative capacity) and exhibit a consensual mode of interaction. By contrast, weakly integrated networks, are typically larger in size, have low barriers to entry for new participants, lack embedded institutional or procedural infrastructure (low administrative capacity) and are defined by a competitive mode of interaction.

In national-level MSF applications, highly integrated networks have been shown to generate fewer policy ideas, which are typically subjected to significant softening-up but are more likely to result in successful policy reform. Conversely, weakly integrated networks tend to generate a larger number of, and more radical, ideas, which, however, have a lower likelihood of leading to adopted reforms due to high levels of competition (Herweg et al., 2018; Zahariadis, 2003).

In the EU’s multi-level governance setting, evaluating policy network integration can provide valuable insights into the nature of successful policy ideas and the suitability of policy entrepreneurship strategies. As highlighted previously, European integration scholarship has often favoured broad, all-encompassing explanations for EU-level policy change, typically emphasising either normative or transactional gains. By contrast, the EU-MSF suggests that such motivations are dynamic and depend, to a large extent, on the level of network integration within the policy area under study.

Specifically, in highly integrated policy networks, where established EU-level procedural and institutional infrastructure facilitates systematic stakeholder interaction, there is less ambiguity regarding technical feasibility. Meanwhile, the consensual mode of deliberation and, typically, the presence of EU-level mediators of national and international preferences reinforce the normative priorities of the network. As a result, it is proposed that achieving structural policy change is more likely to hinge on reconfiguring value acceptability. On the other hand, in weakly integrated EU policy networks, where stakeholders are dispersed across levels of government and interaction is sporadic, it is proposed that technical feasibility is more likely to underpin convergence in policy preferences, as consensus on value acceptability may be nearly impossible to achieve.

Research Design

To answer the research question *Why was an expansive EU HTA framework institutionalised in 2021, after more than three decades of only incremental developments in HTA collaboration at the EU level?* this study employs a process-tracing design, guided by the EU-MSF and informed by original primary data.

As highlighted in the introduction, Regulation (EU) 2021/2282 emerged within an underdeveloped EU-level policy landscape, defined by voluntary collaborative initiatives between national HTA authorities. In

the absence of dedicated EU-level processes and venues, and with fluid stakeholder participation – as will be elaborated in the analysis – the EU-level HTA policy network was weakly integrated.

Against this backdrop, the case allows for testing two hypotheses that link policy network integration to the ripening of the problem and policy streams while assessing the capacity of the EU-MSF to explain the drivers of policy change. On the one hand, with both indicator monitoring and policy feedback occurring primarily in national arenas, it is expected that attention mobilization will not be achieved until EU-level horizontal dissemination is triggered – for example by a focusing event or at the initiative of EU-level actors – even when problems have been long-identified. On the other hand, as elaborated previously, it is hypothesised that in the absence of EU-level administrative capacity, and with competitive and sporadic stakeholder interaction, technical feasibility is likely to be more decisive than value acceptability in enabling transformative policy alternatives to become policy.

The two hypotheses are summarised as follows:

H1: In weakly integrated EU policy networks, achieving attention mobilisation requires the activation of horizontal attention-mobilisation mechanisms, i.e., those that communicate problems across EU-level policy venues.

H2: In weakly integrated EU policy networks, successful policy alternatives depend more heavily on technical feasibility than on value acceptability.

While applying the EU-MSF and testing the two hypotheses, the analysis also examines the exercise of policy entrepreneurship – the connective tissue that enables the coupling of the three streams across all MSF accounts.

The analysis draws on primary sources in the form of framework-driven document analysis and 40 elite interviews with stakeholders active in the policy area during the reform period. The document analysis encompasses binding and non-binding EU acts, Commission strategies, roadmaps and impact assessments, contributions to public consultations, opinions from European Parliament and Council committees, EUnetHTA reports, and outputs from national HTA agencies. It also includes position papers, opinions, annual reports, and press releases from non-institutional stakeholders such as international organisations and pharmaceutical policy interest groups.

Interviews were conducted between January 2022 and June 2023, with the participation of EU policymakers and non-institutional policy stakeholders (Table 1), as selected by a combination of purposive and snowball sampling (response rate: 53%) (Parker et al., 2019; Etikan et al., 2016). They were semi-structured, guided by the analytical axes of the EU-MSF, lasted an average of 30 minutes and were conducted both online and in-person. Table 1 presents a breakdown of the interview participants by affiliation¹.

¹ For the purposes of this study, interviewees are explicitly identified by their role and affiliation only when direct quotations are used.

Table 1: Distribution of Interviewees by Organisational Affiliation	
Affiliation	Number of Participants
European Commission (DG SANTE)	6
Patients' Associations	5
Associations of Pharmaceutical Manufacturers	6
Public Affairs Representatives of the Pharmaceutical Industry	1
European Parliament (Representatives and Support Staff)	5
European Council (Representatives, Permanent Representations and Committee Staff)	5
European Medicines Agency (EMA)	4
EUnetHTA	2
Representatives from National Medicines' Agencies and National HTA Committees of EU Member States	5
World Health Organisation (WHO)	1

Table 1: Distribution of Interviewees by Organisational Affiliation

The EU-level HTA Policy Network (1993-2021)

Following the establishment of the first European HTA body in Sweden in 1987, most EU Member States set up national HTA agencies during the 1990s and 2000s, recognising the value of expert technology assessment in supporting more informed pricing and reimbursement decisions (Banta et al., 2009). Yet, while pharmaceutical policy was undergoing significant EU-level reform in the areas of market authorisation and innovation incentives – marked by the creation of the EMA and the adoption of the General Pharmaceutical Legislation² – no comparable trajectory emerged in the field of HTA.

Instead, EU-level policymaking on HTA was characterised by a series of voluntary collaborative initiatives, driven by national HTA bodies as early as in 1993 and supported initially by the European Commission and, since 2009, also by Member State funding. The European Network for Health Technology Assessment (EUnetHTA) completed three Joint Action (JA) projects between 2009 and 2021, each progressively broader in scope. By JA3, the network operated with a dedicated secretariat and IT platform, was supported by a stakeholder network, had developed joint scientific tools, and produced several joint health technology evaluations and consultations (EUnetHTA, 2012; 2015; 2020a).

Nonetheless, participation and uptake in the joint work remained voluntary, and the network's operation remained dependent on fixed-term funding. At the same time, HTA was identified as a political priority for the European Commission only in 2004 (European Commission, 2004) and institutionalised as an area for cooperation between EU Member States only in 2011 (Directive 2011/24/EU, Article 15).

Amid these conditions, by 2018 – when the Commission submitted its ambitious proposal to overhaul HTA in the EU – the policy network remained weakly integrated. In the absence of a dedicated EU-level institutional venue, it was dispersed across multiple levels of governance, with the locus of power resting in national arenas. Participants – though without consistent presence – included EU institutional actors, national policymakers in health and pharmaceutical policy, HTA evaluators, medical and insurance experts, academics, representatives of the pharmaceutical, generics, and medical devices industries, patient associations, and others. At the same time, the procedural infrastructure for EU policymaking was rudimentary, and EUnetHTA's continuation depended on successive funding renewals.

The Problem Stream

The collaborative efforts of HTA agencies at the EU level contributed to a cumulative process of problem identification from the 1990s onward. However, HTA was not established as a pressing EU-level policy issue until the latter half of the 2010s. During that period, neither the scope nor the nature of identified problems changed significantly. What shifted, instead, were the mechanisms of attention mobilisation that elevated these problems to prominence.

In the EU-MSF, mechanisms of attention mobilisation are grounded in the institutional and procedural interdependencies which manifest within the EU's policymaking system. As such, mechanisms of upward

² The regulatory framework for obtaining approval to develop and market medicines within the EU Single Market (Regulation 726/2004 and Directive 2001/83/EC)

attention mobilisation, whether through policy feedback or indicator reporting, typically involve contributions from Member States, either as part of commitments to EU-level monitoring instruments or through unilateral initiatives in the Council or other fora. Similarly, horizontal attention mobilisation relies on the probing functions of EU supranational institutions as well as the voice of interest associations. In this way, the framework acknowledges and incorporates the multi-level governance dimension of EU policymaking. When issue identification is triggered by domestic actors – such as an expert body, interest group, or independent agency – issue expansion will either aim at national governments, which may then deploy upward mobilisation strategies to Europeanise the problem, or target EU-level interest representatives, which may, in turn, seek to trigger horizontal attention mobilisation mechanisms.

In the case of HTA, while dedicated Member State agencies were established throughout the 1990s and 2000s, governments followed different timelines in allocating resources and determining the mechanisms of HTA uptake in national decision-making. As a result, “for years, most HTA agencies lacked the institutional influence required to lobby Health Ministers and convince them to prioritise HTA issues in Brussels” (*European Commission official*). Moreover, the absence of both a dedicated interest representation network and formal institutional recognition at the EU level placed significant constraints on the activation of horizontal attention-mobilisation mechanisms. For example, the pharmaceutical innovation industry – historically the sector’s most powerful interest group (Permanand & Mossialos, 2005) – initially resisted the emergence of HTA, mounting organised opposition to the additional administrative barriers it introduced to pricing and reimbursement (Oliver et al., 2004). Meanwhile, patient representation in Brussels took decades to develop, and “the contribution of HTA towards better patient outcomes was less emphasised compared to cost-effectiveness” during the early years of its expansion (*Patients’ Association representative*).

In this context, EUnetHTA developed into a forum where domestic policy feedback and, to a lesser extent, both domestic and EU-level indicator monitoring were aggregated and translated into EU-level problems, though without capturing the attention of the wider policy network beyond its own silo.

Attention mobilisation for HTA at the EU level was achieved between 2014 and 2016, in the aftermath of a systemic focusing event and with the catalytic contribution of the European Commission in facilitating EU-level dialogue. The European Sovereign Debt Crisis (or Eurozone crisis) had triggered a prolonged economic downturn in Eurozone Member States from 2009 to the mid-2010s. The resulting recession and austerity measures produced “a combination of decreased public investment and increased consumer burden” that “magnified socioeconomic inequalities in access to treatments” and strained the sustainability of national health systems (*Permanent Representation official*). In response to these challenges, EU Member States called for “more consistent approaches to HTA” as a means of enabling “evidence-based, sustainable, and equitable choices in healthcare and health technologies for the benefit of patients,” as expressed in the 2014 Conclusions of the Employment, Social Policy, Health and Consumer Affairs Council (EPSCO) (Council of the European Union, 2014).

Responding to this call, the European Commission – together with EU-level interest associations – brought to light the challenges facing HTA in the EU policy arena, many of which had already been prominently

highlighted in EUnetHTA outputs over the years. This process of horizontal attention mobilisation was underpinned by the 2016 Roadmap/Inception Impact Assessment on “Strengthening EU Cooperation on Health Technology Assessment (HTA),” launched by the Commission with a distinctly exploratory scope and accompanied by extensive stakeholder consultations (European Commission, 2016). Ultimately, five core problems were identified that had long permeated the EU’s experience with HTA.

First, the European Commission focused on the *adverse effects of duplication*, i.e., the repetition of HTA assessments for the same medicinal products across Member States. “HTAs are multidimensional, and many elements are reasonably controversial” (*Pharmaceutical Industry representative*). To enhance the transferability of assessments and encourage transactional gains, EUnetHTA stakeholders developed the HTA Core Model as early as 2006, a framework that identifies the key domains of HTA (EUnetHTA, 2009). The HTA Core Model distinguishes between the technical and clinical elements of health technology evaluations and their economic, social, legal, and ethical aspects. The former are the focus of a narrower version of HTA, known as the Rapid Relative Effectiveness Assessment (REA) (EUnetHTA, 2012), which, “in theory, should be where most of us [HTA Agencies] can agree, given that we evaluate the same technologies within a Single Market setting” (*National HTA Agency representative*).

Nevertheless, despite the increasing use of the HTA Core Model as a foundation for joint assessments, national-level uptake remained consistently low; below 20% and often involving significant adaptations (EUnetHTA, 2018). At the same time, each individual evaluation at the national level was estimated to cost agencies between €30,000 and €100,000, and the industry between €100,000 and €200,000 (Ecorys, 2013).

Furthermore, the Roadmap/Inception Impact Assessment highlighted significant variation among national HTA settings in terms of the *duration and number of required processes, the scope of assessments* (i.e., which health technologies are included), and *the number of HTAs conducted annually*. Special emphasis was placed on *medical devices and in vitro diagnostics*, an emerging industry that remained subject to minimal HTA coverage in most Member States (European Commission, 2018b).

Beyond variation in processes, substantial differences were also identified in the *methodologies* employed by EU HTA agencies during both Full HTAs and REAs. “The variety in data requirements for the industry and the divergent outcomes of evaluations” often delayed, restricted (through higher prices), or altogether prevented patients’ access to new health technologies (European Commission, 2016). For several Member States, preserving methodological divergence was viewed as a means to prevent a “race to the bottom in terms of evaluation standards” (*National HTA Agency representative*). Nevertheless, the relative importance attributed to different HTA components, and the selection of supplementary material were repeatedly cited as barriers to market access across Member States (Beck et al., 2019).

Confirming H1, despite decades of problem accumulation, the attention of EU policymakers on HTA was not successfully mobilised until 2016, when the Commission, in the aftermath of the Eurozone crisis and in response to the Council’s concerns, aggregated and disseminated the identified problems through the Roadmap/Inception Impact Assessment on HTA. Owing to the weakly integrated status of the policy

network, horizontal mechanisms proved essential to ripening the problem stream for structural policy change.

The Policy Stream – *Recoil-tactics* Policy Entrepreneurship

When a force is applied to a system, such as an explosion inside the chamber of a gun, internal elements are propelled and stretched as far as the exerted force pushes them. However, following the action comes a reaction, pulling the system back toward equilibrium; though now at a different point, shaped by the initial force. This process, known as recoil, resembles the Commission's approach to its ambitious and disruptive proposal for a new EU HTA framework in 2018 (European Commission, 2018a). Within the weakly integrated network, with rudimentary EU-level institutional infrastructure, the Commission succeeded in driving relevant stakeholders to simultaneously communicate their preferences for future policy developments. In doing so, it established a range of technically feasible policy alternatives and proposed an ambitious reform – closely aligned with its own preferences – confident that the ensuing policy deliberation would still result in a transformative shift despite potential pushback.

Central to this process – just as in the problem stream – was the 2016 Roadmap/Inception Impact Assessment on HTA and the accompanying public consultation. In the policy stream, however, what proved instrumental was not simply the initiation of EU-level dialogue but the strategic choices the Commission employed in structuring it. Since the launch of the Better Regulation Agenda (2015), the institutionalisation of public consultations and Roadmaps/Inception Impact Assessments has introduced new dynamics into the EU agenda-setting process (OECD, 2022; Bunea & Ibenskas, 2017; European Commission, 2015). While the Commission is formally obligated to conduct preliminary impact evaluations – comparing the likely effects of policy options and inviting stakeholder input – it retains considerable discretion over both the content and scope of these activities, a flexibility that provides a distinct strategic advantage in policy entrepreneurship. As such, the Commission may communicate a clear policy orientation and gauge its acceptance among affected stakeholders. However, the Commission may also adopt a more tentative approach – one aimed at identifying challenges that may or may not require an EU-level response, informing interested stakeholders of potential policy options and their associated costs, and systematically assessing the balance of policy preferences.

In the case of the new EU HTA framework, the Commission followed the latter course of action. The 2016 Roadmap/Inception Impact Assessment on HTA adopted an exploratory scope, listing all conceivable policy options for the future of EU HTA cooperation – from maintaining the status-quo to establishing a comprehensive, centralised system encompassing both REAs and Full HTAs. The policy options are summarised in Table 2 (European Commission, 2016).

Table 2: Policy Options for HTA as identified in the European Commission's Roadmap/Inception Impact Assessment, 2016							
Policy Option	Description	Financing	Intervention	Common Tools	REAs	Full HTAs	Early Dialogues
1: Status Quo	HTA is regulated and organised at the national/regional level. Following the expiry of the third Joint Action in 2020, and assuming no further action is taken (baseline scenario), there will be no EU cooperation scheme in place afterward.	Current model, until 2020: Jointly funded by the EU Health Programme and Member States	Non-Legislative	Voluntary	Voluntary	Voluntary	Voluntary
2: Long-term voluntary cooperation	Continuation of the current cooperation model, but on a longer-term basis.	Long-term financing mechanism, most likely through the EU Public Health Programme	Non-Legislative	Voluntary	Voluntary	Voluntary	Voluntary
3: Cooperation on collection, sharing and use of common tools and data	Introduction of a legal framework for HTA cooperation, enabling the efforts by national bodies to be compatible, shared and used. Production of joint REA reports on a voluntary basis. Established processes for the use of common tools and early dialogues.	Financing model with EU and Member State contributions as well as industry fees	Legislative	Mandatory	Voluntary	Voluntary	Mandatory
4: Cooperation on the production and uptake of joint REA reports	Member States jointly produce REAs, available to all through a shared repository, with measures for the uptake of the joint work at national level. The assessment of non-clinical domains would remain under the responsibility of Member States.	Permanent/continuous support from the EU, including for a supporting organisational structure. Additionally, Member State contributions and industry fees	Legislative	Mandatory	Mandatory	Voluntary	Mandatory
5: Cooperation on the production and uptake of joint Full HTA reports	Joint production of Full HTA reports	Permanent/continuous support from the EU, Member State contributions and industry fees. Costs likely higher than option 4	Legislative	Mandatory	Mandatory	Mandatory	Mandatory

Table 2: Policy Options for HTA as identified in the European Commission's Roadmap/Inception Impact Assessment, 2016

Moreover, the Commission opted to pair the Roadmap/Inception Impact Assessment with a structured public consultation. Policy stakeholders interested in the future of HTA policymaking in the EU submitted their contributions via a dedicated questionnaire, providing their views on the relative importance of the identified problems and the attractiveness of the proposed policy options³ (European Commission, 2017). This prompted the simultaneous communication of policy preferences by participants in the weakly integrated policy network, both directly and indirectly, in a structured format.

A comparative breakdown of public consultation responses on the usefulness of European HTA outputs revealed high levels of support for joint tools, guidelines, early dialogues, and REAs from both patients' associations and the pharmaceutical innovation industry; the sector's leading interest groups (Figure 1). Notably, patients' preferences extended to joint Full HTAs, whereas the pharmaceutical industry maintained a more reserved stance. Meanwhile, representatives of national HTA agencies, health ministries, and other relevant public organisations also endorsed the usefulness of most joint outputs, but echoed the pharmaceutical industry's reservations regarding Full HTAs. By contrast, representatives of the medical devices industry – traditionally less subject to HTA – expressed significantly lower levels of support for all joint outputs, including both REAs and Full HTAs, than any other stakeholder group.

³ The public consultation on HTA amassed 249 responses including 63 from citizens, 97 from pharmaceutical industry representatives, 27 from public administration representatives, 24 from patients and consumers representatives, 16 from healthcare providers, 8 from academics, 5 from payers and 9 from participants with other relevant roles (European Commission, 2017).

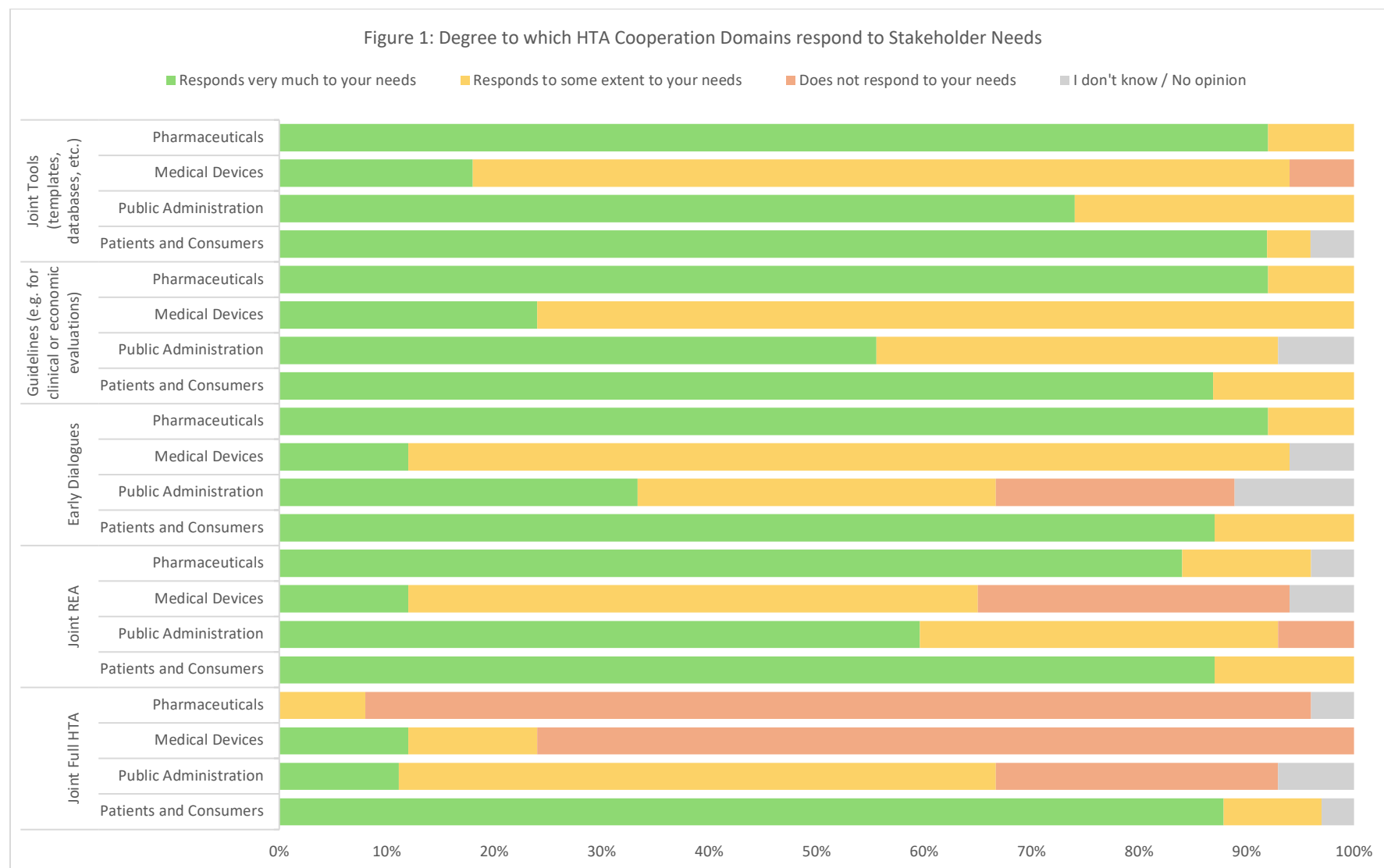


Figure 1: Degree to which HTA Cooperation Domains respond to Stakeholder Needs

Crucially, contributions to the dialogue extended the public consultation inputs. Table 3 summarises the policy preferences of all participants in the EU-level policy network for HTA vis-à-vis joint assessments following the 2016 Roadmap/Inception Impact Assessment.

The EP joined patients' associations, the pharmaceutical innovation industry, and HTA agencies in advocating for a system of joint REAs with mandatory uptake (European Parliament, 2017). For most stakeholders, mandatory participation in joint outputs was of secondary importance, as they recognised the limited capacity and resources of several European HTA agencies. However, the mandatory uptake of REAs was viewed as necessary to justify the investment of resources into joint work.

Patients and the pharmaceutical innovation industry, in particular, favoured “single-dossier REAs” to “streamline the pricing and reimbursement process” and “enhance safety and transparency across Member States”, respectively (*Pharmaceutical Industry and Patients' Association representatives*). National HTA bodies were more divided on the issue, as more resourceful agencies, such as HAS⁴ (France), expressed concerns that mandatory uptake could lead to “the adoption of the lowest common denominator in terms of quality of work” (HAS, 2017). As such, while they accepted a system of mandatory uptake, they insisted on retaining flexibility for national-level adaptations.

This latter approach also proved most acceptable to the Council, which, excluding the medical devices industry, expressed the strongest resistance to a structural transformation of the HTA regime in the EU. As summarised by a *Coreper representative* during the interviews, “Member State positions on HTA were dependent on the trade-off between financial and transactional gains on the one hand and proportionality and subsidiarity concerns on the other”. For countries where the former outweighed the latter – usually due to less robust pricing and reimbursement systems – binding harmonisation was deemed favourable, and vice versa. Nevertheless, even among supporters, there were explicit calls to guarantee the possibility for “national adaptations of joint work” (Finnish Ministry of Health, 2017).

⁴ Haute Autorité de Santé

Table 3: Policy Network Preferences for the Framework of HTA Cooperation beyond 2020

Actor/Institution ID	Policy Alternatives									
	Non-Legislative Continuation of HTA Cooperation	Voluntary Participation in REA	Voluntary Uptake of REA	Mandatory Participation in REA	Mandatory Uptake of REA with National Adaptations	Mandatory Uptake of REA	Voluntary Participation in Full HTA	Mandatory Participation in Full HTA	Mandatory Uptake of Full HTA with National Adaptations	Mandatory Uptake of Full HTA
HTA Agencies	Low Preference	High Preference	Moderate Preference	Low Preference	High Preference	Moderate Preference	Moderate Preference	Low Preference	Low Preference	Low Preference
Pharmaceuticals	Low Preference	High Preference	Low Preference	Low Preference	Moderate Preference	High Preference	Low Preference	Low Preference	Low Preference	Low Preference
Medical Devices	High Preference	Low Preference	Low Preference	Low Preference	Low Preference	Low Preference	Low Preference	Low Preference	Low Preference	Low Preference
Patients	Low Preference	Low Preference	Low Preference	High Preference	Moderate Preference	High Preference	Moderate Preference	High Preference	High Preference	High Preference
Council	Moderate Preference	High Preference	Moderate Preference	Low Preference	Moderate Preference	Low Preference	Low Preference	Low Preference	Low Preference	Low Preference
European Parliament	Low Preference	Moderate Preference	Moderate Preference	High Preference	High Preference	High Preference	Low Preference	Low Preference	Low Preference	Low Preference

Table 3: Policy Network Preferences for the Framework of HTA Cooperation beyond 2020

Therefore, by 2017 – without yet having submitted an official proposal for reform – the Commission had successfully leveraged the public consultation to pre-emptively identify the policy preferences of key stakeholders within the weakly integrated network, despite the absence of institutional infrastructure for systematic stakeholder interaction. By mass-communicating problems that had accumulated within EUnetHTA forums over decades and prompting policy actors to deliberate on all conceivable policy options, the Commission was able to chart a range of feasible policy alternatives (Figure 2).

Confirming H2, no policy option exhibited consensus-level value acceptability; however, stakeholders did converge in their assessments of technical feasibility, which in turn primarily shaped their policy preferences (for example, the pharmaceutical industry’s support for mandatory uptake in joint REAs, but not in Full HTAs). As such, the Commission could confidently deem some degree of institutionalised cooperation on HTA, centred around joint REAs, as technically feasible, with uptake emerging as the most controversial dimension when shifting from more voluntary to more mandatory models. Therefore, it could propose any policy option within this range of feasibility, knowing that even after likely pushback, the resulting “recoil” would still constitute substantial policy change, breaking with decades of incrementalism.

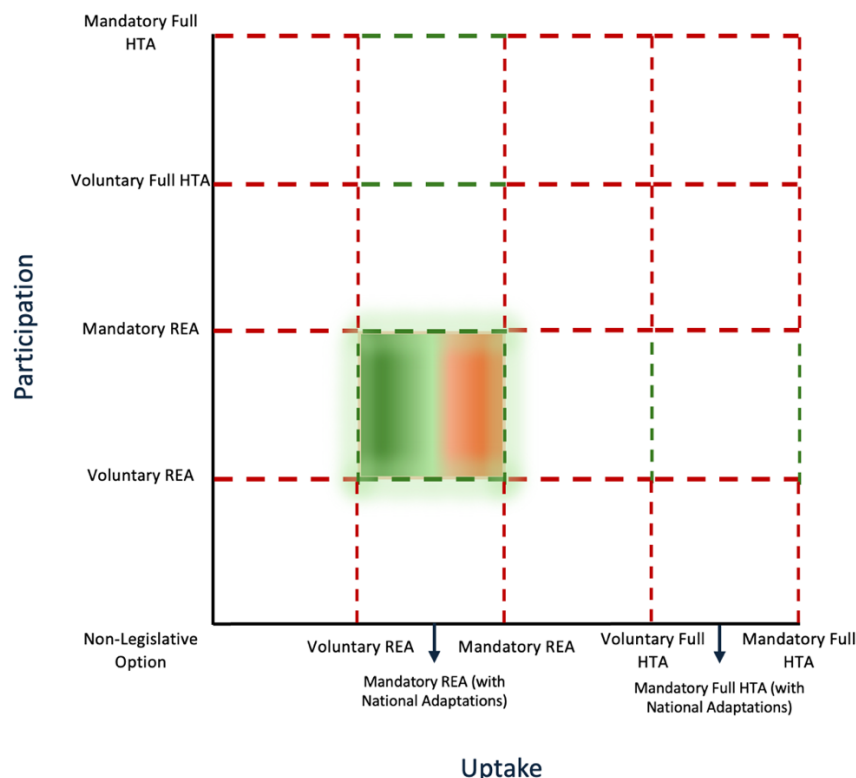


Figure 2: Range of Feasible Alternatives for EU HTA Reform (2016-2018)

The Politics Stream

On 31 January 2018, a year after the conclusion of the public consultation, the European Commission submitted a *Proposal for a Regulation of the European Parliament and of the Council on Health Technology Assessment and Amending Directive 2011/24/EU* (European Commission, 2018a). The Commission's proposal stood in full alignment with the 'functional' perception of the EU public that it typically serves.

First, and most radically, the Commission proposed a transition to joint clinical assessments (REAs) for all medicinal products undergoing the central market authorisation procedure, as well as for high-risk medical devices and in vitro diagnostics. The Commission's 'functional' narrative echoed the discourse of patients' associations, the pharmaceutical industry, and HTA decision-makers, promoting EU-level consistency and transparency as substantial improvements to the evidence base for pricing and reimbursement processes. Beyond JCAs, the Commission's proposal also provided for the conduct of Joint Scientific Consultations (JSCs) and annual horizon scanning for emerging health technologies. The executive responsibilities for JCAs, JSCs, and horizon scanning would be delegated to a new EU-level instrument, the 'Coordination Group', made up of national HTA agency representatives appointed by Member States (European Commission, 2018a).

Responding to the Commission's proposal, the European Parliament showed strong support for the outlined policy direction, while introducing amendments to frame the new EU HTA framework in line with the 'popular' perception of the EU public it typically serves. Most notably, the EP complemented Article 114, originally selected by the Commission as the legal basis for the reform, with Article 168(4) of the TFEU, framing the HTA Regulation not only in relation to internal market improvements but also to public health protection. As such, the EP emphasised unmet medical need considerations in the selection of medical technologies for JCA and the evaluation of JSC requests. Moreover, the EP enhanced the legislative provisions on transparency and accountability – including conflict of interest measures – to promote public scrutiny of JCAs and JSCs, and to frame the reform as an improvement in decision-making pluralism and fairness.

Last, the Council converged on a policy position by 2021, tailoring the proposal to the interests of a 'ruling' perception of the EU public. For the more sceptical Member States, the proposal's "very far-reaching measures" were seen as a threat to the "tried and tested procedure for testing medicinal products" (German Bundestag, 2018). As such, they conflicted with the narrative of empowering national health systems to make safe and efficient decisions by reducing regulatory dissonances within the internal market.

Consequently, the Council's amendments to Regulation (EU) 2021/2282 focused on considerations of necessity and on reinforcing Member States' ownership of domestic HTA outcomes. First, the mandate for developing the JCA and JSC methodologies was transferred from the Commission to the Member States. Second, an incremental implementation timeline was introduced, beginning with cancer medicines and high-risk medical devices, and expanding to all medicinal products by 2030, reaffirming the importance of technical feasibility, as proposed in H2. As noted by a *European Commission representative*, "at the current stage, we are not looking at anything more than 10 JCAs for the first year." From a narrative standpoint,

the incremental implementation approach also enabled the prioritisation of the most expensive medical technologies – those most likely to be unattainable for patients in several Member States.

Last, and most notably, the obligations for mandatory uptake were reduced. The Council successfully advocated for Member States to retain the ability to supplement or revisit JCA conclusions, while requiring them to give due consideration to the published JCA reports, annex them to domestic assessments, and refrain from requesting clinical data or other evidence already submitted at the EU level. Upon the conclusion of the decision-making process, the Council’s more reserved political determination effectively constrained the initial ambition of the reform. Nevertheless, despite the expected pushback, the legislative process still delivered the most significant structural reform of HTA to date, thereby crowning the Commission’s *recoil-tactics* policy entrepreneurship with success (Figure 3).

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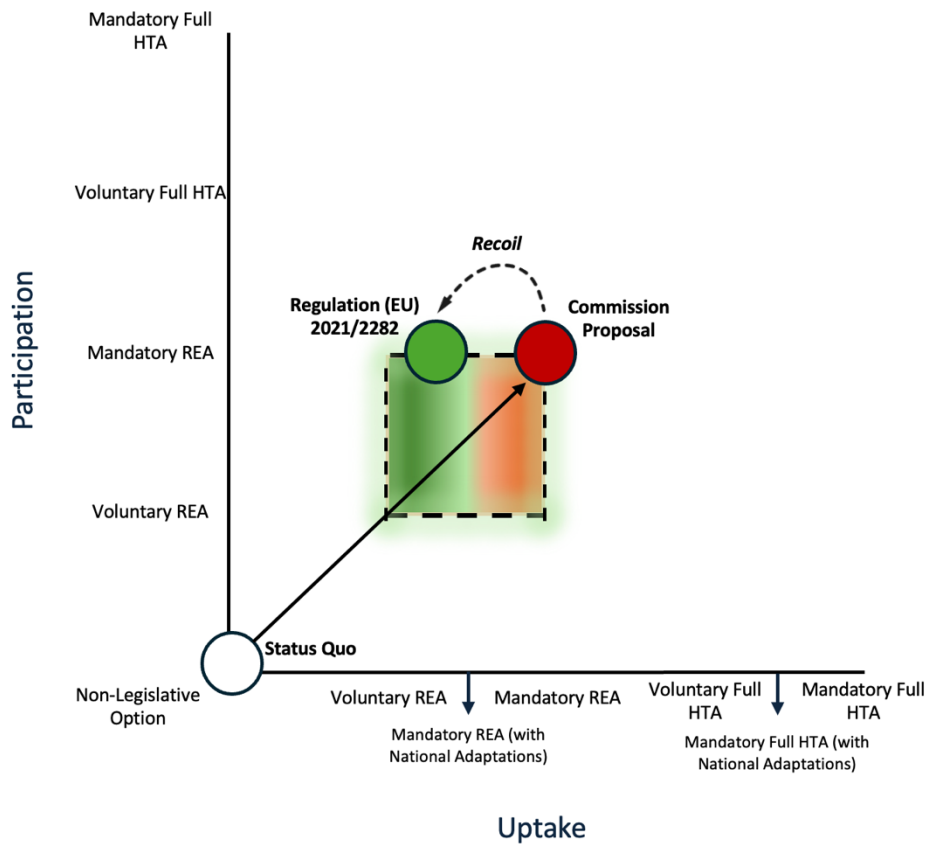


Figure 3: 'Recoil-tactics' Policy Entrepreneurship in the case of the new EU HTA Framework (Regulation (EU) 2021/2282)

Appendix Table 1 provides a detailed overview of the evolution of the legislative process for the new EU HTA framework

Discussion – Conclusions

Ultimately, *why was an expansive EU HTA framework institutionalised in 2021 after more than three decades of only incremental developments in HTA collaboration at the EU level?* And how do this paper's conclusions inform the EU policy process and MSF scholarship, as well as the pharmaceutical policy literature?

The process-tracing account employed in this study, guided by the EU-MSF, identified *recoil-tactics* policy entrepreneurship – used by the European Commission between 2016 and 2018 – as integral to driving structural EU-level reform for HTA. More than two decades ago, Susanne Schmidt, in her seminal work, questioned whether the Commission should “only” be viewed as “an agenda-setter” within the EU's political system (Schmidt, 2000). Schmidt's argument, developed in a markedly different EU decision-making reality before the signing of the Treaty of Lisbon, suggested that the Commission, acting in its role as “Guardian of the Treaties,” could manipulate voting outcomes in the Council of Ministers. In today's EU policymaking environment, dominated by the Ordinary Legislative Procedure, the use of *recoil-tactics* demonstrates how the Commission can influence both agenda and decision outcomes within a more fluid and pluralistic system, by strategically leveraging preference ambiguity.

The use of *recoil-tactics* involves “stretching” the EU-MSF's streams in order to induce policy actors to simultaneously publicise policy concerns and preferences. Using these inputs to navigate policymaking ambiguity, the policy entrepreneur – in this case, the European Commission – can establish a range of feasible policy alternatives, propose the one most aligned with its narrative, and credibly expect that interinstitutional bargaining will converge on an option within that range. In this way, the Commission can induce a policy outcome that still represents a preferential, and likely structural, departure from the status quo, like Regulation (EU) 2021/2282.

Central to enabling this strategic approach was the weakly integrated nature of the policy network. After three decades of largely voluntary collaboration at the EU level, the field of HTA was marked by long-recognised problems – such as the administrative and financial burdens caused by duplication, and the pricing and reimbursement delays and inconsistencies arising from divergent evaluation methodologies across Member States. However, in the absence of established EU institutional venues, these problems failed to mobilise the attention of policymakers and stakeholders beyond the EUnetHTA circles. In this context, and as confirmed by H1, horizontal attention-mobilisation mechanisms that emerged in the aftermath of the Eurozone crisis – most notably the 2016 Roadmap/Inception Impact Assessment, which surfaced EUnetHTA inputs and enabled organised interests to communicate policy feedback – proved integral in ripening the problem stream. While the Commission did not act as a problem broker (Knaagard, 2015), it provided the necessary space for others – such as EUnetHTA representatives or patients' associations – to engage in problem brokering.

Moreover, in the policy stream, the low levels of policy network integration impeded any prolonged scrutinisation of policy alternatives. Confirming H2, structural policy change in less integrated networks is more likely to rely on technical feasibility than value acceptability, given the wide network membership

and the lack of mechanisms for softening up ideas. In the face of financial hardship for Member State health systems, the Commission was able to lay all conceivable alternatives for EU HTA on the table, emphasising the administrative and fiscal gains of joint work at the EU level and referencing the experience of EUnetHTA to promote the existence of adequate policy capacity. As a result, despite EU-level HTA collaboration having only grown in small incremental steps in the past, the Commission was able to establish a range of feasible alternatives, effectively drawing from a blank canvas.

Here, a special note must be made on the increasing usefulness of EU Public Consultations for both policymakers and policy researchers. In today's EU policy reality, the Commission is mandated to formally involve interested stakeholders throughout the agenda-setting process. While public consultations have often been viewed as sources of political grandstanding in the EU's bid for democratisation, recent scholarship suggests that they are becoming an indispensable part of the early policymaking stages (Bunea and Lipcean, 2024; OECD, 2022). Crucially, the Commission retains considerable freedom in deciding the scope and format of consultations, obtaining the ability to strategically utilise non-binding legislative acts for the development of future policy proposals.

In the case of Regulation (EU) 2021/2282, the Commission not only outlined a broad range of policy options in the Roadmap/Inception Impact Assessment but also invited stakeholders to contribute structured feedback as part of an exploratory exercise. Ultimately, informing *recoil-tactics* policy entrepreneurship was the comprehensive and systematic expression of actor preferences, supplementing and/or replacing the usual Brussels-based lobbying, which relies more heavily on relative access dynamics. Given the publicly accessible nature of this information, policy scholars can draw unique aggregated conclusions about official stakeholder positions, which can be corroborated with more informal primary data, such as interviews like in this study. Similar approaches can further be employed in comparable forums for preference aggregation in other multi-level governance settings, such as the G7 or the World Trade Organization (WTO).

Last, in the politics stream, interinstitutional bargaining constrained the initial ambition of the reform. Yet institutional narratives converged on a position that still enabled structural policy change – mandating participation in JCAs for all centrally authorised medicinal products and, over time, most categories of medical devices, while also establishing safeguards to ensure substantial uptake.

When policy windows are triggered by the sudden mobilisation of policymakers' attention to emerging problems – for instance, through a focusing event as in the case of HTA – institutional narratives are shaped during or after policy formulation, as opposed to cases when a prior narrative shift opens the window. This logic resembles the distinction between consequential and doctrinal couplings introduced by Zahariadis (2007). In the case of Regulation (EU) 2021/2282, the problem-driven window enabled the Commission to advance the alternative most aligned with its own position without pre-emptively seeking compromise. Ultimately, this may prove highly advantageous policy entrepreneurs, as unlikely instances of narrative convergence – such as between the Commission and the Council on the scope of JCAs – can drive policy outcomes even closer to the original proposal than initially anticipated.

More broadly, the approach and findings of this study highlight the analytical capacity of the EU-MSF to explain EU-level policy change across sectors and initiatives. The framework's tools – such as directional attention-mobilisation mechanisms and political narratives – address contextual, analytical, and measurement limitations, enabling the MSF logic to be extended to the EU level and to guide a systematic programme of study. As emphasised in the introductory sections, contemporary EU policy change requires dedicated policy process lenses, and the MSF provides the most suitable foundation for meeting this challenge. Moreover, the hypotheses tested and confirmed in this study invite further evaluation – either in comparable cases or, conversely, in policy initiatives that emerge within highly integrated networks. Through such applications, the EU-MSF can be systematically refined, much as MSF scholarship has evolved in Western democracies, specific regions (e.g., the United States or Europe), the subnational level, and, more recently, in autocracies (see Zahariadis et al., 2023 for relevant representative examples).

Finally, the paper's findings are highly relevant to the EU pharmaceutical policy literature. The new EU HTA framework officially entered into force on January 12, 2025, following the conclusion of the transitional period. This study offers the first – and highly timely – account of the drivers behind this transformative shift, complementing previous research on the early stages of the agenda-setting process (see, most notably, Loblova, 2021).

Importantly, the analysis highlights not only the regulatory changes but also the evolving policy process dynamics that produced them. The Europeanisation of the HTA problem after decades of marginalisation, the systematic engagement of key interest groups (including the pharmaceutical industry and patients' associations), the establishment of a dedicated institutional venue, and the construction of procedural infrastructure together create a new policymaking reality for HTA. These developments will shape both the framework's implementation trajectory and future reforms – for example, the potential extension of HTA to vaccines. Similar dynamics are visible across other areas of EU pharmaceutical policy, which has been undergoing significant reform since the latter half of the 2010s. These include the revision of the General Pharmaceutical Legislation and the establishment of the European Health Union – both of which underscore the need for further policy process research in this sector.

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All EU Legislative Acts were accessed through EUR Lex [<https://eur-lex.europa.eu>]

Directive 2001/83/EC

Directive 2011/24/EU

Regulation (EC) No 726/2004

Regulation (EU) 2021/2282

Appendix

Appendix Table 1: Comparative Overview of the Commission's Proposal, the EP's Amendments and the Final Version of Regulation (EU) 2021/2282			
Legislative Provisions	European Commission	European Parliament	Council of the European Union (Regulation Final Draft)
Governance Mechanism	Member State Coordination Group: Executive and strategic responsibilities, including an Annual Work Programme and Annual Report Comprised of national authorities and bodies responsible for health technology assessment, appointed by Member States on an ad-hoc or permanent basis Decisions by consensus, or, where necessary, by simple majority Dedicated sub-groups for joint clinical assessments, joint scientific consultations, identification of emerging health technologies, voluntary cooperation, preparation of the annual work programmes and annual reports, and updates of the common rules and working documents Support Framework: <u>Commission:</u> Responsibility to host, oversee and co-chair the meetings of the Coordination Group, provide secretariat, administrative, scientific and IT support, and facilitate cooperation between the Coordination Group, the EMA and relevant Union bodies on medical devices <u>Stakeholder Network:</u> open call and selection process, informational exchanges with the Coordination Group, may nominate clinical experts and patients to attend meetings of the Coordination Group as observers	Member State Coordination Group: Executive and strategic responsibilities, including an Annual Work Programme and Annual Report Comprised of national and regional authorities and bodies responsible for health technology assessment, without any financial interests in any type of health technology developer industry or insurance company that may affect their impartiality, appointed by Member States on an ad-hoc or permanent basis Decisions by consensus, or, where necessary, by qualified majority Dedicated sub-groups for joint clinical assessments, joint scientific consultations, identification of emerging health technologies, voluntary cooperation, preparation of the annual work programmes and annual reports, and updates of the common rules and working documents Support Framework: <u>Commission:</u> Responsibility to host, oversee and co-chair the meetings of the Coordination Group, without the right to vote, provide secretariat, administrative, scientific and IT support, and facilitate cooperation between the Coordination Group, the EMA and relevant Union bodies on medical devices <u>Stakeholder Network:</u> open call and selection process addressed at patient associations, consumer organisations, non-governmental organisations in the field of health, health technology developers and health professionals. The European Parliament shall have two representatives. Responsibilities include informational exchanges with the Coordination Group, feedback on the preparation of the Annual Work Programme and nominating clinical experts and patients to attend meetings of the Coordination Group as observers	Member State Coordination Group: Executive and strategic responsibilities, including an Annual Work Programme and Annual Report Comprised of national and regional authorities and bodies responsible for health technology assessment, without any financial interests in any type of health technology developer industry or insurance company that may affect their impartiality, appointed by Member States on an ad-hoc or permanent basis Decisions by consensus, or, where necessary, by qualified majority for the adoption of the Annual Work Programme, Annual Report and Strategic Direction Decisions by consensus, or where necessary by simple majority for matters of technical or scientific nature (e.g., JCAs, JSCs etc.) Dedicated sub-groups for joint clinical assessments, joint scientific consultations, identification of emerging health technologies, development of methodological and procedural guidance. Support Framework: <u>Commission:</u> Responsibility to host and oversee the meetings of the Coordination Group, provide secretariat, administrative, scientific and IT support, decide of conflict of interest and facilitate cooperation between the Coordination Group, the EMA and relevant Union bodies on medical devices <u>Stakeholder Network:</u> open call and selection process addressed at all eligible organisations, in particular patient associations, consumer organisations, non-governmental organisations in the field of health, health technology developers and health professionals. Responsibilities include informational exchanges with the Coordination Group and receiving updates on the progress of joint work. The Coordination Group may invite members of the stakeholder network to attend its meetings as observers.

Joint Clinical Assessments (REAs):	Scope: All medicinal products undergoing the central marketing authorisation procedure (Regulation (EC) No 726/2004) Medical devices classified as class IIb and III (highest risk) according to Regulation (EU) 2017/745 and <i>in vitro</i> diagnostic medical devices classified as class D (highest risk) according to Regulation (EU) 2017/746 for which the relevant expert panels have provided a scientific opinion Participation: Voluntary, for a three-year transition period Mandatory, thereafter Uptake: Voluntary, for a three-year transition period Mandatory, thereafter Methodology: A common procedural and methodological framework for clinical assessments, procedures for joint clinical assessments and procedures for joint scientific consultations, including distinct rules for medicinal products and medical devices, established by the Commission Timeline: Synchronised with market authorisation decision	Scope: All medicinal products undergoing the central marketing authorisation procedure (Regulation (EC) No 726/2004), and products not covered by the regulation but where the health technology developer has opted for the centralised authorisation procedure, and they constitute a major technical, scientific or therapeutic innovation. Medical devices classified as class IIb and III (highest risk) according to Regulation (EU) 2017/745 and <i>in vitro</i> diagnostic medical devices classified as class D (highest risk) according to Regulation (EU) 2017/746 for which the relevant expert panels have provided a scientific opinion and considered to be a significant innovation and with potential significant impact on public health or health care systems Participation: Voluntary, for a three-year transition period Mandatory, thereafter Uptake: Voluntary, for a three-year transition period Mandatory, thereafter Although not allowed to duplicate, Member States can consider additional clinical data and evidence when necessary to complete the health technology assessment or the overall pricing and reimbursement process, with the obligation to inform the Coordination Group of these intentions and justify the choice. Methodology: A common procedural and methodological framework for clinical assessments, procedures for joint clinical assessments and procedures for joint scientific consultations, including distinct rules for medicinal products and medical devices, established by the Coordination Group. In the case of orphan medicines, there may be determined that there is no substantive reason to support further clinical analysis beyond the significant benefit assessment already carried by the EMA. Timeline: Synchronised with market authorisation decision	Scope: All medicinal products undergoing the central marketing authorisation procedure (Regulation (EC) No 726/2004) and authorised medicinal products for which a joint clinical assessment report has been already published but a new therapeutic indication is introduced. Medical devices classified as class IIb and III (highest risk) according to Regulation (EU) 2017/745 and <i>in vitro</i> diagnostic medical devices classified as class D (highest risk) according to Regulation (EU) 2017/746 for which the relevant expert panels have provided a scientific opinion. Participation: Phased mandatory participation: From January 2025, for oncology and advanced therapy medicinal products (ATMPs) and high-risk medical devices and in vitro diagnostics From January 2028, for orphan medicinal products From January 2030, for all other centrally authorised medicinal products Uptake: Mandatory obligation to: 1. Give due consideration to the published joint clinical assessment reports 2. Annex the dossier submitted by the health technology developer to the documentation of the HTA at Member State level 3. Annex the published joint clinical assessment report to the HTA report at Member State level 4. Not request at the national level information, data, analyses or other evidence that has been submitted by the health technology developer at Union level 5. Immediately share through the IT platform referred any information, data, analyses and other evidence with the Coordination Group that they receive from the health technology developer at Member State level Methodology: A common procedural and methodological framework for clinical assessments, procedures for joint clinical assessments and procedures for joint scientific consultations, including distinct rules for medicinal products and medical devices, established by the Coordination Group Timeline: Synchronised with market authorisation decision
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Joint Scientific Consultations:	Scope: Health technology developers may request a joint scientific consultation with the Coordination Group for the purposes of obtaining scientific advice concerning data and evidence likely to be required as part of a joint clinical assessment Selection: The Coordination Group shall evaluate joint scientific consultation requests according to the following criteria: 1. the likelihood that the health technology under development will be the subject of a joint clinical assessment 2. unmet medical needs 3. potential impact on patients, public health, or healthcare systems 4. significant cross-border dimension 5. major Union-wide added value 6. the available resources Timeline: The Coordination Group shall inform health technology developers on the outcome of their request within 15 days and prepare a report within 100 days. Health technology developers may request that the joint scientific consultation takes place in parallel with the process of receiving scientific advice from the European Medicines Agency for market authorisation.	Scope: Health technology developers may request a joint scientific consultation with the Coordination Group for the purposes of obtaining scientific advice concerning the optimal design of studies to obtain the best scientific evidence, improve predictability, align research priorities and enhance the quality and efficiency of said research, to obtain the best evidence. Selection: The Coordination Group shall evaluate joint scientific consultation requests according to the following criteria: 1. the likelihood that the health technology under development will be the subject of a joint clinical assessment 2. unmet medical needs 3. potential impact on patients, public health, or healthcare systems 4. significant cross-border dimension 5. major Union-wide added value 6. the available resources 7. Union clinical research priorities Timeline: The Coordination Group shall inform health technology developers on the outcome of their request within 15 days and prepare a report within 100 days. Health technology developers may request that the joint scientific consultation takes place in parallel with the process of receiving scientific advice from the European Medicines Agency for market authorisation.	Scope: The Coordination Group shall carry out joint scientific consultations in order to exchange information with health technology developers on their development plans for a given health technology. Those consultations shall facilitate the generation of evidence that meets the likely evidence requirements of a subsequent joint clinical assessment on that health technology Selection: The Coordination Group shall evaluate joint scientific consultation requests according to the following criteria: 1. unmet medical needs 2. first in class 3. potential impact on patients, public health, or healthcare systems 4. significant cross-border dimension 5. major Union-wide added value 6. Union clinical research priorities Timeline: The Coordination Group shall inform health technology developers on the outcome of their request within 15 days. The Commission shall send the joint scientific consultation outcome document to the requesting health technology developer at the latest 10 working days after it has been finalised. JSCs on medicinal products may take place in parallel with the process of receiving scientific advice from the European Medicines Agency for market authorisation. JSCs on medical devices may take place in parallel with the consultation of the expert panels pursuant to Article 61(2) of Regulation (EU) 2017/745.
Emerging Health Technologies (Horizon Scanning)	Scope: Annual study prepared by the Coordination Group on emerging health technologies expected to have a major impact on patients, public health or healthcare systems. The conclusions of the study shall be summarised in the Coordination Group's annual report and be taken into account in the preparation of its annual work programmes. Process: Consultative process between the Coordination Group and health technology developers, patients' organisations, clinical experts, the EMA, the Medical Devices Coordination Group (Regulation (EU) 2017/745)	Scope: Annual study prepared by the Coordination Group on emerging health technologies expected to have a major impact on patients, public health or healthcare systems. The conclusions of the study shall be summarised in the Coordination Group's annual report and be taken into account in the preparation of its annual work programmes. Process: Consultative process between the Coordination Group and health technology developers, patient and consumer organisations, health professionals, organisations, clinical experts, the EMA, the Medical Devices Coordination Group (Regulation (EU) 2017/745)	Scope: The Coordination Group shall ensure the preparation of reports on emerging health technologies expected to have a major impact on patients, public health or healthcare systems. Reports shall in particular address the estimated clinical impact and the potential organisational and financial consequences of emerging health technologies for national healthcare systems. Process: The preparation of the reports shall be based on existing scientific reports or initiatives on emerging health technologies and information from relevant sources, including clinical study registers and scientific reports, the EMA, the Medical Device Coordination Group, health technology developers and members of the stakeholder network. The Coordination Group may consult

			stakeholder organisations which are not members of the stakeholder network and other relevant experts, as appropriate.
Voluntary Cooperation on Health Technology Assessment	Scope: The Commission shall support cooperation and the exchange of scientific information among Member States on: 1. non-clinical assessments on health technologies 2. collaborative assessments on medical devices 3. health technology assessments on health technologies other than medicinal products or medical devices 4. the provision of additional evidence necessary to support health technology assessments The Coordination Group shall be used to facilitate the cooperation	Scope: The Commission shall support any further cooperation and the exchange of scientific information among Member States on: 1. non-clinical assessments on health technologies 2. collaborative assessments on medical devices 3. health technology assessments on health technologies other than medicinal products or medical devices 4. the provision of additional evidence necessary to support health technology assessments 5. clinical assessments of medicinal products and medical devices carried out by Member States 6. measures relating to compassionate use in clinical practice in order to improve the evidence basis and to create a register for this purpose 7. the development of best medical practice guides based on scientific evidence 8. disinvestment in obsolete technologies 9. the tightening of the rules on clinical evidence generation and its monitoring The Coordination Group shall be used to facilitate the cooperation	Scope: The Commission shall support cooperation and the exchange of scientific information among Member States on: 1. non-clinical assessments on health technologies 2. collaborative assessments on medical devices and in vitro diagnostic medical devices 3. HTAs on health technologies other than medicinal products, medical devices or in vitro diagnostic medical devices 4. the provision of additional evidence necessary to support HTAs, in particular in relation to health technologies for compassionate use and obsolete health technologies 5. emerging health technologies with a major anticipated impact on patients, public health or healthcare systems. The Coordination Group shall be used to facilitate the cooperation
Funding	Union Funding for the work of the Coordination Group and sub-groups Assessor and co-assessors shall be entitled to a special allowance compensating them for their work on joint clinical assessments and joint scientific consultations	Union Funding for the work of the Coordination Group and sub-groups, stable and permanent, without the direct or indirect funding by developers of health technologies The Commission may establish a system of charges for health technology developers requesting both joint scientific consultations and joint clinical assessments which it shall use to finance research regarding unmet medical needs or clinical priorities. Such a system of charges shall under no circumstances used to finance activities under this Regulation. Assessor and co-assessors shall be entitled to a special allowance compensating them for their work on joint clinical assessments and joint scientific consultations	Union Funding for the work of the Coordination Group and sub-groups in support of the work on joint clinical assessments and joint scientific consultations, including the development of methodological guidance, and on the identification of emerging health technologies. Assessor and co-assessors shall be entitled to a special allowance compensating them for their work on joint clinical assessments and joint scientific consultations.

Appendix Table 1: Comparative Overview of the Commission's Proposal, the EP's Amendments and the Final Version of Regulation (EU) 2021/2282